Deep-Dive Explanation — cam_info_tf_check.py

What the node does

This ROS 2 node is a sanity checker for two things:

1) that your CameraInfo is publishing valid intrinsics; and

2) that a TF transform exists from a camera optical frame to a chosen base/world frame.

It subscribes to a sensor_msgs/CameraInfo topic, prints the intrinsics once (fx, fy, cx, cy), and then every second

tries to look up the transform target ← source (defaults: platform_base ← cam2_color_optical_frame). If available,

it logs translation and quaternion; otherwise, it warns until TF shows up.

Parameters (and the important gotcha)

- info_topic (default: /camera/cam2/color/image_raw/theora) — this should be your camera's .../camera_info topic.

  The default in the script points to a video topic; override it at runtime.

- source_frame (default: cam2_color_optical_frame) — camera optical frame (child).

- target_frame (default: platform_base) — base/world frame (parent).

Subscriptions, TF, and timers

• CameraInfo subscription: created with QoSProfile(depth=10). Camera drivers often publish with SENSOR_DATA QoS;

  if you see no messages, switch to QoSPresetProfiles.SENSOR_DATA.

• TF: a tf2_ros.Buffer + TransformListener are set up. A 1.0 s timer calls lookup_transform(target, source, Time()).

  Passing Time() requests the latest transform. On success it prints t=(x,y,z) and q=(x,y,z,w); on failure it warns.

How intrinsics are extracted

On the first CameraInfo, the node reads the pinhole K matrix:

  fx = K[0], fy = K[4], cx = K[2], cy = K[5] and logs them with the incoming frame_id. That confirms that the camera model is wired correctly.

Log output you should expect

When CameraInfo arrives:

CameraInfo received (frame_id=cam2_color_optical_frame) fx=... fy=... cx=... cy=...

Then every second (if TF exists):

TF OK platform_base <- cam2_color_optical_frame | t=(0.123,0.456,1.234) q=(0.000,0.707,0.000,0.707)

Otherwise:

TF not available yet: <error details>

Note the arrow target ← source: you're asking for the transform from source_frame to target_frame.

Quick improvements (optional but recommended)

- Use SENSOR_DATA QoS for CameraInfo:

sensor_qos = QoSPresetProfiles.SENSOR_DATA.value

self.info_sub = self.create_subscription(CameraInfo, info_topic, self.on_info, sensor_qos)

- Fix the default info_topic to your actual .../camera_info.
- Exit-on-timeout mode: add a parameter to stop with non-zero status if TF never appears (nice for CI scripts).